

Heme is found in several proteins, including hemoglobin in blood and myoglobin in muscle. It has a four-coordinate iron-bound porphyrin structure where ferrous iron (Fe^{2+}) is bound to four nitrogen atoms through coordinate covalent bonds. The most common heme found in the body is heme *b* in hemoglobin (Figure 1).

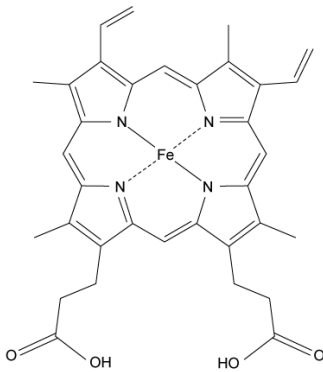


Figure 1 Structure of heme *b* found in hemoglobin

Along with the four nitrogen ligands in the porphyrin structure, His F8 also binds to Fe^{2+} in the axial position, leaving the other axial position available to bind oxygen. In this setting, iron's atomic *d* orbitals are no longer degenerate. The energetic difference between the *d* orbitals depends on the ligands.

In deoxygenated heme, iron sits out of the plane formed by the nitrogen ligands in the porphyrin because its *d* orbitals make it slightly too large to fit in the porphyrin ring. However, when O_2 binds to heme, iron's electrons rearrange themselves: the *d* orbitals become smaller, and iron sits within the plane of the nitrogen atoms.

When iron is scarce in the body, Zn^{2+} coordinates to the ligands in the protoporphyrin structure through a nonenzymatic process, forming Zn-protoporphyrin IX (ZPP). ZPP and protoporphyrin IX (PPIX) have distinct emission spectra in the blue range when excited, whereas heme does not. Hematofluorometry can be used as a simple, inexpensive indicator for iron deficiency in whole blood by placing a blood sample in a fluorometer and measuring the amount of ZPP relative to the amount of heme.

Figure 2 illustrates the fluorescence emission spectra for ZPP and PPIX measured from a blood sample when ZPP is excited at 425 nm and PPIX is excited at 407 nm.

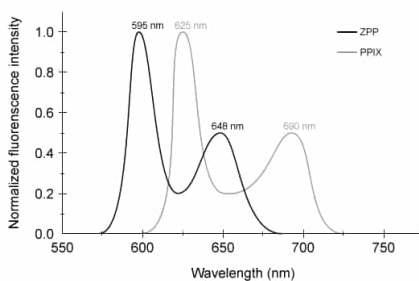


Figure 2 ZPP and PPIX fluorescence emission spectra

What is the electron configuration of the iron cation that binds with the porphyrin ligand to form heme *b*?

- ☐ A. $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^6$ [13%]
- ☐ B. $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^4$ [47%]
- ☒ C. $1s^2 2s^2 2p^6 3s^2 3p^6 3d^6$ [34%]
- ☐ D. $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^8$ [5%]

Correct

34% answered correctly

Explanation:

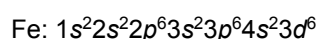
Electron configurations in the periodic table

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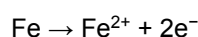
An **electron configuration** accounts for the placement of all electrons within the **shells and subshells** of an atom or ion. Shells are indicated using the **principal quantum number** n , which relates to the size of the shell and its distance from the nucleus. Subshells are labeled by type as s , p , d , or f and hold a maximum of 2, 6, 10, and 14 electrons, respectively. The subshell types indicate the **shapes of the orbitals** that a subshell contains. Each orbital is associated with a different energy level, and the **Aufbau principle** states that electrons fill these levels in **order** of increasing energy from lowest to highest.

The **periodic table** is arranged into **subshell blocks** based on the type of orbital occupied in the **valence shell**. With respect to the valence shells of the elements, the s orbitals are on the two left columns (except helium), the p orbitals are on the six right columns, the d orbitals are in the middle, and the f orbitals are at the bottom. The subshell blocks **indicate the electron configuration** for a selected element based on its position on the periodic table. Starting at hydrogen, one electron is added to the appropriate orbital for each square crossed (moving from left to right across each row) until the chosen element is reached.

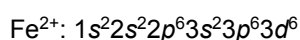
Following this method, the electron configuration for a neutral Fe atom (26 protons, 26 electrons) is:



In their elemental state, atoms contain an equal number of protons and electrons (ie, **no net charge**). If an **ion is formed**, the charge of the ion is determined by how many electrons are gained or lost. If electrons are lost, the electrons in the highest **valence shell** (ie, the shell with the largest value of n) are removed first. An Fe^{2+} cation forms when Fe loses two electrons:



As such, the electrons lost by the Fe atom come from the highest shell ($n = 4$), so the $4s$ electrons are removed. Therefore, the electron configuration of the iron cation that binds with the porphyrin ligand to form heme *b* is:



(Choice A) This electron configuration is for a neutral Fe atom rather than an iron cation.

(Choice B) Compared to the electron configuration of Fe, this configuration shows two electrons removed from the $3d$ sublevel, but electrons are generally lost first from the shell that is farthest from the nucleus (with the highest value of n).

(Choice D) Compared to the electron configuration of Fe, this configuration shows two electrons added to the $3d$ sublevel; however, cations are formed by losing electrons.

Educational objective:

The Aufbau principle states that lower-energy orbitals are filled first. Subshell blocks on the periodic table can be used as a general guide to orbital filling to determine the electron configuration of an atom or ion. Each block on the table represents the different orbital types in the valence shells of the elements the block contains, with s orbitals in the two left columns, p orbitals in the six right columns, d orbitals in the middle, and f orbitals at the bottom.

Time spent: 0 sec
Subject:
General Chemistry

QID: 400968
System:
Atomic Theory and Chemical Composition

Heme is found in several proteins, including hemoglobin in blood and myoglobin in muscle. It has a four-coordinate iron-bound porphyrin structure where ferrous iron (Fe^{2+}) is bound to four nitrogen atoms through coordinate covalent bonds. The most common heme found in the body is heme *b* in hemoglobin (Figure 1).

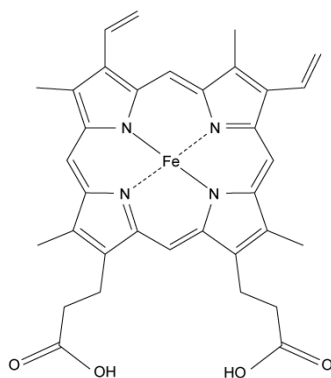


Figure 1 Structure of heme *b* found in hemoglobin

Along with the four nitrogen ligands in the porphyrin structure, His F8 also binds to Fe^{2+} in the axial position, leaving the other axial position available to bind oxygen. In this setting, iron's atomic *d* orbitals are no longer degenerate. The energetic difference between the *d* orbitals depends on the ligands.

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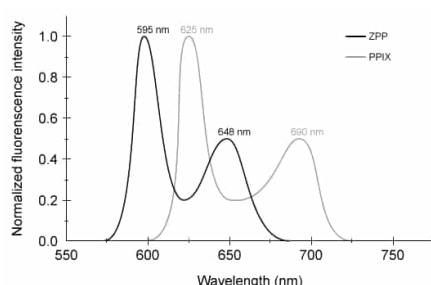


Figure 2 ZPP and PPIX fluorescence emission spectra

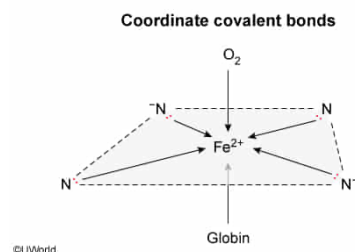
Which of the following statements does NOT describe the iron-porphyrin complex of heme *b* formed between Fe^{2+} and the four nitrogen atoms in Figure 1?

- ☐ A. Each nitrogen atom provides both bonding electrons in each coordinate bond. [23%]
- ☒ B. The net charge of the iron-porphyrin complex is +2. [38%]
- ☐ C. Nitrogen's electrons interact with iron's *d* orbitals. [17%]
- ☐ D. Iron's electron configuration determines coordinate bond strength. [20%]

Correct

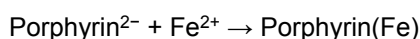
37% answered correctly

Explanation:



Coordinate covalent bonds are a special type of bond formed from a **Lewis acid-base interaction** between a **central atom**, such as a metal cation, and a **ligand**. Both electrons in a coordinate bond come from the ligand. Essentially, the ligands share their lone pairs of electrons with vacant orbitals in the metal cation, but the ligands maintain possession of their electrons and the metal ion maintains its oxidation state. The metal and its ligands together form a **coordination complex**, in which the number of coordinate bonds to the metal cation is known as the **coordination number**.

To determine the formal charges of a metal cation and its ligands prior to forming a complex, a thought experiment can show that the electrons in the coordinate bonds are returned to the ligands. Doing this for the iron-porphyrin complex from Figure 1 shows that the porphyrin ligand must have a net charge of -2 before binding:



The Fe^{2+} cation is positively charged but two of the donor nitrogen atoms in the porphyrin ligand both have two bonds and two lone pairs, giving them each a -1 formal charge. These negatively charged nitrogen atoms form the $\text{N}-\text{Fe}$ coordinate bonds. The other two donor nitrogen atoms in the porphyrin ligand each have three bonds and one lone pair, causing them both to be neutral (uncharged). These neutral nitrogen atoms form the $\text{N}\cdots\text{Fe}$ coordinations but their formal charges are unchanged because the electrons are only loaned to the vacant orbitals of the metal cation.

Therefore, after binding the resulting net charge of the iron-porphyrin complex is zero (not $+2$):

$$\text{Net charge} = \text{N}^- + \text{N}^- + \text{N}^0 + \text{N}^0 + \text{Fe}^{2+} = -1 - 1 + 0 + 0 + 2 = 0$$

(Choice A) The nitrogen atoms in the porphyrin ligand each have a lone pair of electrons that provide both bonding electrons in each coordinate covalent bond.

(Choice C) The lone pairs of nitrogen's electrons interact with iron's d orbitals to form the coordinate bond. In a coordinate bond, the d orbitals are no longer degenerate (ie, are not equal in energy) and have distinct shapes, some of which point directly at the negatively charged ligands.

(Choice D) The strength of the coordination bond depends on which d orbitals have electrons in them.

Educational objective:

Coordinate covalent bonds are a special bond formed from a Lewis acid-base interaction between a central atom and its ligands. The ligands provide bonding electrons, which interact with the metal's d orbitals to form the coordinate covalent bond, and the number of coordinate bonds is the coordination number.

Time spent: 0 sec
Subject:
General Chemistry

QID: 400969
System:
Atomic Theory and Chemical Composition

Heme is found in several proteins, including hemoglobin in blood and myoglobin in muscle. It has a four-coordinate iron-bound porphyrin structure where ferrous iron (Fe^{2+}) is bound to four nitrogen atoms through coordinate covalent bonds. The most common heme found in the body is heme *b* in hemoglobin (Figure 1).

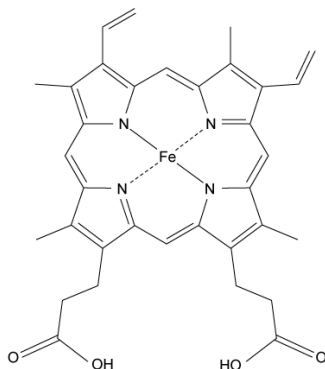


Figure 1 Structure of heme *b* found in hemoglobin

Along with the four nitrogen ligands in the porphyrin structure, His F8 also binds to Fe^{2+} in the axial position, leaving the other axial position available to bind oxygen. In this setting, iron's atomic *d* orbitals are no longer degenerate. The energetic difference between the *d* orbitals depends on the ligands.

In deoxygenated heme, iron sits out of the plane formed by the nitrogen ligands in the porphyrin because its *d* orbitals make it slightly too large to fit in the porphyrin ring. However, when O_2 binds to heme, iron's electrons rearrange themselves: the *d* orbitals become smaller, and iron sits within the plane of the nitrogen atoms.

When iron is scarce in the body, Zn^{2+} coordinates to the ligands in the protoporphyrin structure through a nonenzymatic process, forming Zn-protoporphyrin IX (ZPP). ZPP and protoporphyrin IX (PPIX) have distinct emission spectra in the blue range when excited, whereas heme does not. Hematofluorometry can be used as a simple, inexpensive indicator for iron deficiency in whole blood by placing a blood sample in a fluorometer and measuring the amount of ZPP relative to the amount of heme.

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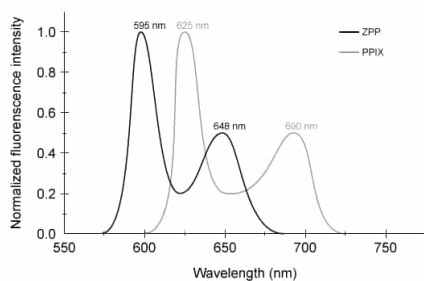
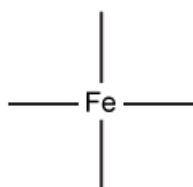


Figure 2 ZPP and PPIX fluorescence emission spectra

Which of the following depicts iron's coordination geometry when heme *b* is bound to O_2 and His F8 in hemoglobin?

☐ A.



[15%]

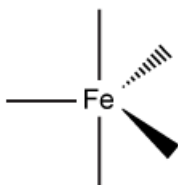
☐ B.





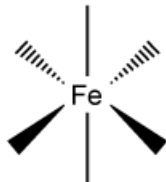
[9%]

☐ C.



[9%]

☒ D.

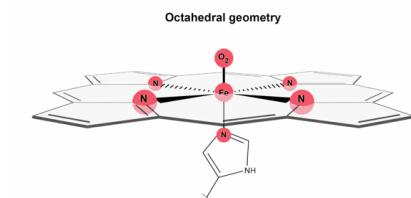


[64%]

Correct

64% answered correctly

Explanation:



Like **covalent bond geometry** in valence shell electron pair repulsion (VSEPR) theory, coordinate covalent bonds **maximize geometric space** around the **central atom** to minimize repulsion between electrons. Transition metals tend to form complexes with two, four, or six coordinate bonds, which means the **coordination geometry** around the central atom is typically **linear**, **tetrahedral**, **square planar**, or **octahedral**. Ligands capable of forming more than one coordinate bond occupy more than one bond position around the metal center geometry.

As described in the passage, when heme *b* (Figure 1) axially binds with O_2 and a globin histidine, Fe^{2+} has three ligands (ie, porphyrin, histidine, O_2) that together form six **coordinate covalent bonds**: four to the nitrogen atoms in the porphyrin ring, one to histidine from the globin protein, and one to an O_2 molecule. Therefore, Fe^{2+} and its ligands form an octahedral arrangement in which the six coordinate bonds are all at approximately 90° angles from each other.

(Choices A and B) These structures display planar and tetrahedral geometry, respectively, and therefore both indicate only four coordinate bonds. The iron in oxygenated hemoglobin has six coordinate bonds.

(Choice C) This structure has trigonal bipyramidal geometry with only five coordinate bonds around the central atom. The iron in oxygenated hemoglobin has six coordinate bonds.

Educational objective:

Coordinate covalent bonds tend to maximize geometric space around a central atom. In general, transition metals form complexes with two, four, or six coordinate bonds. Therefore, the geometry of the coordinate covalent bonds is typically linear (two bonds), tetrahedral or square planar (four bonds), or octahedral (six bonds).

Heme is found in several proteins, including hemoglobin in blood and myoglobin in muscle. It has a four-coordinate iron-bound porphyrin structure where ferrous iron (Fe^{2+}) is bound to four nitrogen atoms through coordinate covalent bonds. The most common heme found in the body is heme *b* in hemoglobin (Figure 1).

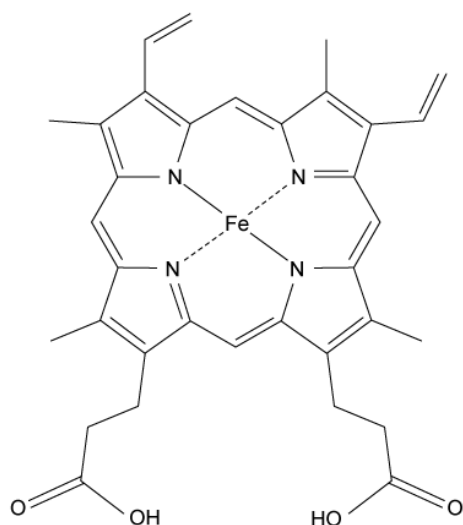


Figure 1 Structure of heme *b* found in hemoglobin

Along with the four nitrogen ligands in the porphyrin structure, His F8 also binds to Fe^{2+} in the axial position, leaving the other axial position available to bind oxygen. In this setting, iron's atomic *d* orbitals are no longer degenerate. The energetic difference between the *d* orbitals depends on the ligands.

In deoxygenated heme, iron sits out of the plane formed by the nitrogen ligands in the porphyrin because its *d* orbitals make it slightly too large to fit in the porphyrin ring. However, when O_2 binds to heme, iron's electrons rearrange themselves: the *d* orbitals become smaller, and iron sits within the plane of the nitrogen atoms.

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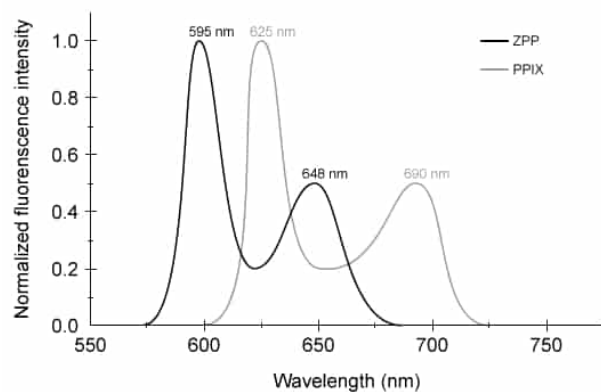
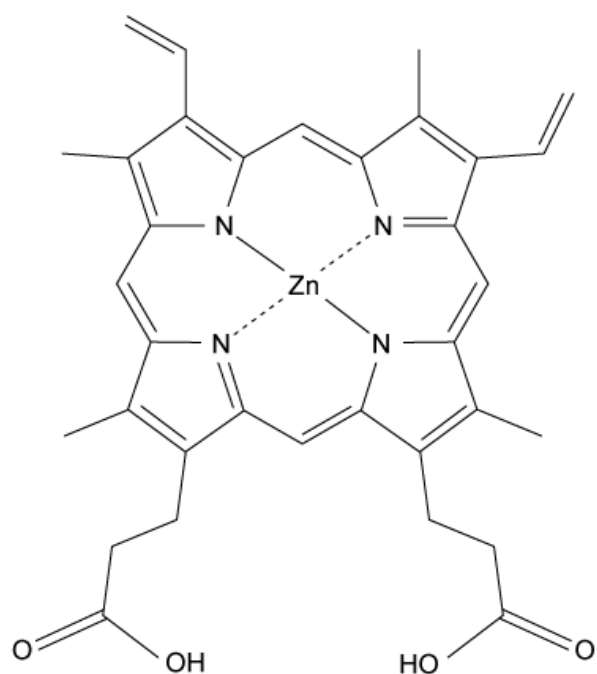


Figure 2 ZPP and PPIX fluorescence emission spectra

The figure below shows Zn^{2+} substituted in place of Fe^{2+} in heme *b* to form ZPP, which cannot bind to His F8 or O_2 . Based on the structure of ZPP, which of the following is the coordination number for Zn^{2+} ?



- ☐ A. 2 [17%]
- ✓ ☒ B. 4 [73%]
- ☐ C. 6 [5%]
- ☐ D. The coordination number cannot be determined from the information given. [3%]

Correct

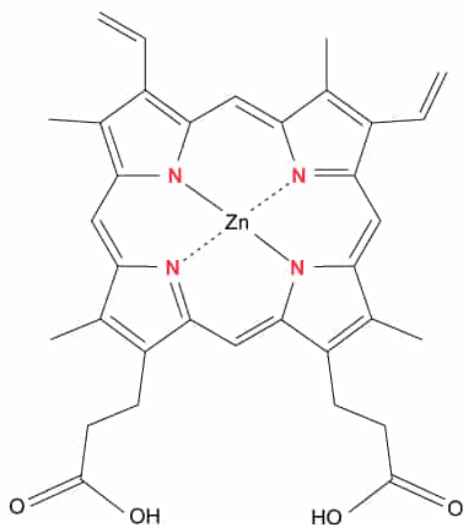
73% answered correctly

Explanation:

Explanation:

Coordination number

Coordination number = number of coordinate bonds = 4



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A **complex ion** is formed when a **central atom**, usually a **metal ion**, is bonded to **ligands** through a **coordinate covalent bond**. The number of coordinate bonds to the central atom is known as the **coordination number**.

The four nitrogen atoms within the cavity of the PPIX ring form coordinate covalent bonds with the central atom. In ZPP, the central atom is Zn^{2+} and it forms four coordinate covalent bonds, making its coordination number 4.

(Choice A) Even though Zn^{2+} has a +2 oxidation state, Zn^{2+} is coordinated to four nitrogen atoms via their lone pairs of electrons. Coordinate covalent bonds do not necessarily balance the charge on the central atoms. This characteristic is different from an ionic bond, such as ZnCl_2 , in which Zn^{2+} bonds to the number of Cl^- ions required to balance its charge.

(Choice C) The central Fe^{2+} atom in heme *b* forms coordinate covalent bonds with the four nitrogen atoms of the PPIX ring as well as with His F8 and O_2 (ie, a coordination number of 6). In contrast, Zn^{2+} cannot bind to His F8 and O_2 and has a coordination number of 4. A metal's preferred coordination number involves several factors, including the size of the metal ion, the number of electrons in the metal's *d* orbitals, and the size of the ligands.

(Choice D) The image of ZPP and the information in the passage provide enough information to know how many coordinate bonds are formed with Zn^{2+} . In ZPP, the

central Zn^{2+} atom prefers a coordination number of 4, so it does not bind to His F8 or to O_2 , but Zn^{2+} does form coordinate covalent bonds with the four nitrogen atoms in the PPIX ring.

Educational objective:

A complex ion consists of a central atom (typically a metal ion) and its surrounding ligands, which form coordinate covalent bonds to the metal. The number of coordinate bonds to the central atom is the coordination number.

Time spent:0 sec

Subject:
General Chemistry

QID:400971

System:
Atomic Theory and Chemical Composition

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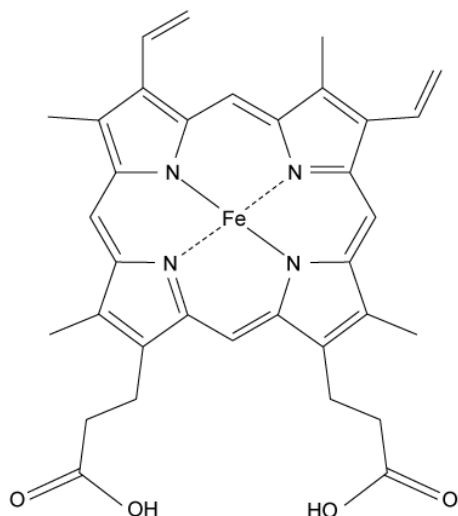


Figure 1 Structure of heme *b* found in hemoglobin

Along with the four nitrogen ligands in the porphyrin structure, His F8 also binds to Fe^{2+} in the axial position, leaving the other axial position available to bind oxygen. In this setting, iron's atomic *d* orbitals are no longer degenerate. The energetic difference between the *d* orbitals depends on the ligands.

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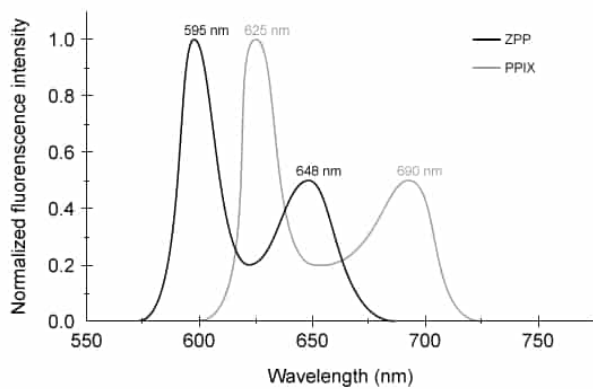


Figure 2 ZPP and PPIX fluorescence emission spectra

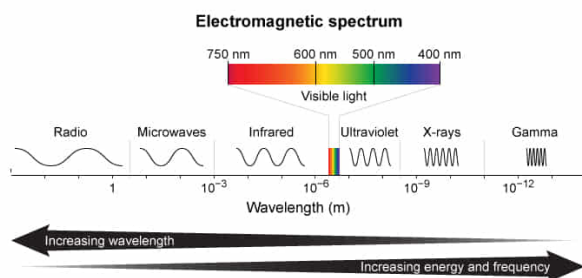
Considering the peak excitation wavelength stated in the passage and the highest-intensity fluorescence peak in Figure 2, which of the following is true about ZPP detection in blood samples?

- ☐ A. ZPP and PPIX have peak photon emissions that are equal in energy. [6%]
- ☒ B. ZPP requires less energy to excite its electrons from the ground state than does PPIX. [38%]
- ☐ C. The energy of the photons used to excite ZPP is less than the energy of the photons emitted by the molecule during fluorescence. [18%]
- ☐ D. The energy of the photons absorbed by ZPP during excitation is greater than the energy change of the resulting electronic transition in the molecule. [35%]

Correct

38% answered correctly

Explanation:



The **energy of a photon** is **inversely proportional** to its **wavelength** (ie, shorter wavelengths correspond to higher energies). Some molecules can **absorb a photon** in a process called **excitation**, causing an electron to move from the ground state to a higher-energy (excited) state with the change in the electron's energy equal to the energy of the absorbed photon. The electron can then relax back to the ground state

by emitting light, heat, or both. When some of the energy is lost as heat and the rest is lost as light, the emitted photon has less energy than the absorbed photon, and therefore has a longer wavelength. This process is known as **fluorescence**.

As stated in the last sentence of the passage, ZPP is excited by photons with a wavelength of 425 nm and PPIX is excited by photons with a wavelength of 407 nm. The photons used to excite ZPP have a *longer* wavelength (ie, lower energy) than the photons used to excite PPIX. Therefore, ZPP requires *less* energy to excite its electrons from the ground state than does PPIX.

(Choice A) The peaks of the emission spectra in Figure 2 are at different wavelengths, so ZPP and PPIX do not have peak photon emissions of the same energy.

(Choice C) Because the wavelength of the photons used to excite ZPP (425 nm) is shorter than the wavelength of its highest-intensity emissions (595 nm) in Figure 2, the energy of the photons used to excite ZPP is *greater than* the energy of the photons emitted by the molecule during fluorescence.

(Choice D) When an electron in a molecule absorbs a photon, the electron transitions from a lower to a higher energy level, gaining energy equal to the energy of the absorbed photon. As such, the energy of a photon absorbed by ZPP is *equal to* the energy change of the resulting electronic transition occurring in the molecule.

Educational objective:

Electromagnetic radiation at a certain wavelength consists of photons of a particular energy, and that energy is inversely proportional to the wavelength. An electronic transition from a lower to a higher energy level occurs when a molecule absorbs a photon that has an energy equal to the energy difference between the two levels.

Time spent:0 sec

Subject:

General Chemistry

QID:400972

System:

Atomic Theory and Chemical Composition

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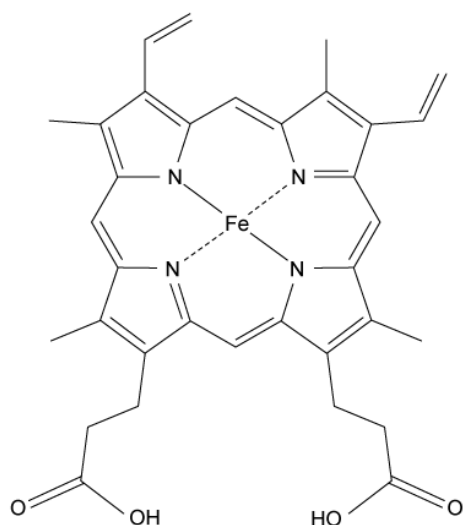


Figure 1 Structure of heme *b* found in hemoglobin

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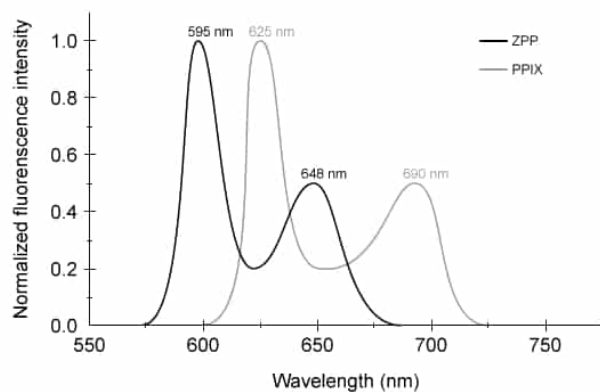


Figure 2 ZPP and PPIX fluorescence emission spectra

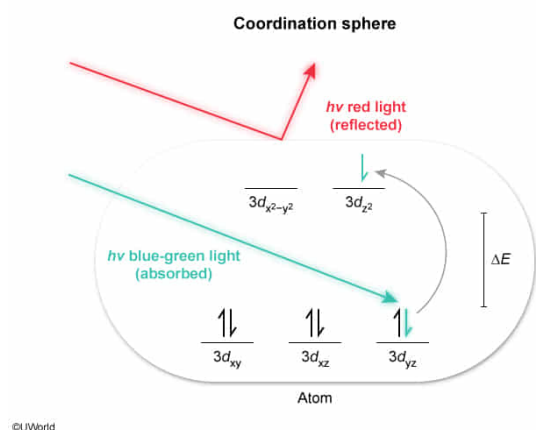
Which of the following statements best explains why hemoglobin is red when it binds to O_2 ?

- ✓ ☒ A. O_2 interacts with iron's d orbitals. [67%]
- ☐ B. O_2 is a nonpolar ligand. [1%]
- ☐ C. O_2 has lone pairs of electrons. [6%]
- ☐ D. O_2 changes the surrounding protein structure. [24%]

Correct

67% answered correctly

Explanation:



When placed within a **coordination sphere**, the atomic **d orbitals** of a metal are **no longer degenerate**, so they have different energies. This occurs because some orbitals point toward the ligands, which are electron donors, and other d orbitals point away from or between the ligands. The orbitals that point toward the ligands are higher in energy than the orbitals pointing away from or between ligands because

electrons repel each other. The **energies** of the orbitals **determine** which wavelengths of light can be absorbed. Some ligands cause greater differences in the energy of the *d* orbitals than other ligands, resulting in the **absorption of different wavelengths of light**. These wavelengths are usually in the **visible spectrum**.

When heme binds O_2 , the nature of the O_2 ligand changes the energy of iron's *d* orbitals. This energy change causes heme to absorb blue-green light and reflect red light.

(Choice B) The polarity of O_2 does not affect color. Color is due to electron transitions.

(Choice C) Lone pairs of electrons on O_2 are a factor in the difference in energy in iron's *d* orbitals. However, the red color is due to electron transitions between the *d* orbitals, not the lone pair on O_2 .

(Choice D) Although hemoglobin's protein conformation does change upon binding O_2 , this change in conformation is not the cause of the red color in blood.

Educational objective:

The nature of the ligands in a coordination sphere causes the metal's *d* orbitals to have different energies. This energy difference determines the wavelength of light absorbed by the bonds. The wavelength of light reflected is often in the visible region of the electromagnetic spectrum.

Time spent:0 sec

Subject:

General Chemistry

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System:

Atomic Theory and Chemical Composition